MIL-E-36700A(DSA-DM)
30 June 1976
SUPERSEDING
MIL-E-36700
23 July 1970

MILITARY SPECIFICATION

ETHOHEPTAZINE CITRATE AND ASPIRIN TABLETS

This specification is approved for use by the Defense Personnel Support Center, Defense Supply Agency, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Ethoheptazine Citrate and Aspirin Tablets (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals. (See 6.3)

STANDARDS

FEDERAL

FED-STD-140 - Tablets (for Medicinal Purposes).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Ethoheptaazine Citrate and Aspirin Tablets, and shall be in accordance with all applicable requirements of FED-STD-140, and as specified herein.

S2. Classification. Shall be type I.

S5.3 The following is added to this paragraph:

Shall be tablets containing 75 mg ethoheptazine citrate and 325 mg aspirin within the designated assay limits for the tablets.

S6.4.9.1 Tablets shall disintegrate in not more than 25 minutes, using Simulated Gastric Fluid T.S., as the immersion fluid.

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3.2 Assay of the finished tablets.

3.2.1 Aspirin. Shall assay to contain not less than 95.0 percent and not more than 110.0 percent of the required amount of aspirin, when determined as specified in 4.3.1.2.

3.2.2 Free salicylic acid. Shall assay to contain not more than 0.20 percent free salicylic acid at the time of manufacture, and not more than 1.50 percent free salicylic acid throughout the shelf life of the tablet, based on the labeled aspirin content, when determined as specified in 4.3.1.1.

3.2.3 Ethoheptazine citrate. Shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amount of ethoheptazine citrate when determined as specified in 4.3.1.3.

3.3 Identification of active ingredients in the finished tablet.

3.3.1 Aspirin. The identification of aspirin shall be determined as specified in 4.4.1.

3.3.2 Ethoheptazine citrate. The identification of ethoheptazine citrate shall be determined as specified in 4.4.2.

3.4 Ingredients.

3.4.1 Aspirin. The aspirin used in the manufacture of the tablets shall comply with the tests, standards, and requirements of the U.S.P., including any supplements or revisions thereto.

3.4.2 Ethoheptazine citrate. The ethoheptazine citrate used in the manufacture of the tablet shall comply with the tests, standards and requirements of the NFX111 which was official from September 1, 1970 to June 30, 1975.

3.5 Other ingredients. All other ingredients used in the manufacture of the tablets shall comply with S5.1 of FED-STD-140.

3.6 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period (potency period) specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated item, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. The sampling as shown in FED-STD-140 shall be applicable.

4.3 Tests.

4.3.1 Finished product. Accurately weigh 20 tablets and determine the average tablet weight. Reduce the tablets to a fine powder in a dry mortar and reserve the powdered sample in a tightly closed container for subsequent assays.

4.3.1.1 Assay for free salicylic acid.

Reagents:

Extraction Solvent - Dilute 100 ml of glacial acetic acid to 2000 ml with ethyl acetate.

Ferric Nitrate Reagent - Dissolve 2 grams of ferric nitrate nonahydrate. $Fe(NO_3)_3.9H_2O$, and 1 ml of nitric acid in purified water and dilute to 200 ml. Further dilute 25 ml of the first solution (stock solution) to 500 ml with purified water to obtain the working reagent.

Standard Salicylic Acid Solutions - Dissolve about 500 mg of Salicylic Acid, U.S.P. accurately weighted, in Extraction Solvent and dilute to 100.0 ml. Further dilute 1.0, 4.0 and 10.0 ml portions of the first dilution (stock solution) to 100 ml with Extraction Solvent to obtain the working solutions. Calculate the exact concentrations of these working solutions, mg salicylic acid per ml.

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Assay Preparation - Transfer a finely-powdered accurately weighed sample equivalent to about 1 gram of aspirin, to a fine-porosity sintered-glass funnel. Attach the funnel to a suction bell jar containing a 100 ml) wide-mouth volumetric flask. Applying suction, extract the sample with four 20 ml portions of Extraction Solvent, stirring the funnel contents occasionally for about one minute before drawing each extract into the volumetric flask. Adjust the final volume of the solution in the flask to 100.0 ml and mix well.

Procedure - To separate Klett tubes (matched) add 1.0 ml portions of the three working Standard Salicylic Acid Solutions, the Assay Preparation, and the Extraction Solvent (blank). To each tube, add 1.0 ml of ethanol (absolute alcohol) and 10.0 ml of Ferric Nitrate Reagent. Invert the tubes to mix thoroughly. (If the sample solution is not clear at this point it may be filtered and collected in another Klett tube.) Concomitantly read the colors of these solutions, using a suitable colorimeter (such as Klett Colorimeter with a #54 filter or equivalent instrument giving comparable results), and using the blank to zero the instrument. Calculate as follows:

$$\frac{Ru}{Rs} \times \frac{100C}{Wu} = \text{mg of free salicylic acid/mg of sample (record this value for information - needed to calculate the aspirin assay)}$$

$$\frac{Ru}{Rx} \times \frac{100C}{Wu} \times \frac{Wav.}{K} \times 100 = \text{percent free salicylic acid, based on labeled aspirin content}$$

where,

Ru = sample reading
Rs = standard reading nearest to the sample reading
C = concentration of working standard solution, mg salicylic acid per ml
Wu = sample weight, mg
Wav. = average weight, mg/tablet
K = labeled aspirin content, mg/tablet

4.3.1.2 Assay for aspirin.

Reagent:

Ferric Nitrate Reagent - Dissolve 2 grams of Ferric Nitrate nonahydrate, $Fe(NO_3)_3.9H_2O$, and 1 ml of nitric acid in purified water and dilute to 200 ml. Further dilute 25 ml of the first solution (stock solution) to 500 ml with purified water to obtain the working reagent.

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Standard Salicylic Acid Solutions - Dissolve about 500 mg of Salicylic Acid U.S.P., accurately weighed, in ethanol (absolute alcohol) and dilute to 100.0 ml. Further dilute 4.0 ml of the first dilution (stock solution) to 100 ml with ethanol to obtain the working solution. Calculate the exact concentration of this working solution (mg salicylic acid per ml).

Assay Preparation - Transfer a finely-powdered accurately weighed sample, equivalent to about 230 mg of aspirin, to a 250 ml of volumetric flask. Add 10.0 ml of 1N sodium hydroxide solution and 90 ml of purified water, then heat the mixture on a steam bath for one hour. Add 11.0 ml of 1N hydrochloric acid, cool to room temperature, and dilute to volume with purified water. Mix and filter the mixture through filter paper, rejecting the first 20 ml of filtrate, and then dilute 10.0 ml of the filtrate to 50.0 ml with purified water.

Procedure - To separate Klett tubes (matched) add the following:

Tube	Standard Salicylic Acid Solution	Assay Prep.	Purified Water	Ethanol	Ferric Nitrate Reagent
sample	---	1.0 ml	----	1.0 ml	10.0 ml
standard	1.0 ml	---	1.0 ml	---	10.0 ml
blank	---	---	1.0 ml	1.0 ml	10.0 ml

Invert the tubes to mix thoroughly, then read the colors of the sample and standard, using a suitable colorimeter (such as Klett Colorimeter with a #54 filter or equivalent instrument giving comparable results), and using the blank to zero the instrument. Calculate as follows:

$$\left[\left(\frac{Ru}{Rs} \times \frac{1250C}{Wu}\right) - \left(\text{mg. of free salicylic acid/mg. of sample}\right)\right] \frac{Wav}{K} \times 130.4 = \text{Percent Labeled amount of aspirin}$$

where, Ru = sample reading
Rs = standard reading
C = concentration of working standard solution, mg salicylic acid per ml
Wu = sample weight, mg
Wav. = average weight, mg/tablet
K = Labeled mg aspirin/tablet

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4.3.1.3 Assay for ethoheptazine citrate.

Reagent:

Perchloric Acid Solution (0.025 N) - Dilute 2 ml of 72 percent perchloric acid to one liter with glacial acetic acid and standardize this solution as follows:

Transfer an accurately weighed sample of about 200 mg of 1,3-diphenylguanidine to a 250 ml wide-mouth Erlenmeyer flask containing a magnetic stirring bar. Add 20 ml glacial acetic acid, 120 ml of carbon tetrachloride and 2 drops of crystal violet T.S. While stirring, titrate with Perchloric Acid Solution until the violet color just changes to blue.

$$\text{normality} = \frac{\text{mg of diphenylguanidine}}{\text{ml of titrant} \times 211.3}$$

Procedure - Transfer about 1.3 grams of powdered tablets, accurately weighed, to a 150 ml separator. Add 20 ml of alkaline salt solution (100 ml of saturated sodium chloride solution plus 2 grams of sodium hydroxide), stopper, and shake vigorously for about 5 minutes. Extract with four 30 ml portions of carbon tetrachloride, filtering the extracts through cotton overlaid with anhydrous sodium sulfate (both cotton and sodium sulfate previously wetted with carbon tetrachloride). Collect the filtered extracts in a wide-mouth 250 ml Erlenmeyer flask containing a magnetic stirring bar. Add 20 ml of glacial acetic acid and 2 drops of crystal violet T.S. to the flask, and titrate with Perchloric Acid Solution until the violet color just changes to blue. Perform a blank titration in the same manner.

Calculation -

$$\frac{(Vu-Vb) \times N \times 0.4535}{Wu} \times \frac{Wav}{K} \times 100 = \text{Percent labeled amount of Ethoheptazine}$$

where, Vu = sample titration, ml
Vb = blank titration, ml
N = exact normality of Perchloric Acid Solution
Wu = sample weight, gram.
Wav = average weight per tablet, mg
K = labeled ethoheptazine citrate per tablet

4.4 Identification of active ingredients in the finished tablet. Using 5 tablets, separate the aspirin from the ethoheptazine citrate and reserve for their respective identifications.

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4.4.1 Aspirin. A violet color shall be produced when the tablets are treated as follows:

Add about 500 mg of the aspirin to 20 ml purified water. Heat on a steam bath for 5 minutes. Filter through No. 42 Whatman Paper, add 3 drops of ferric alum reagent to the filtrate. Note: Ferric Alum reagent may be prepared as follows: To 15 ml of ferric ammonium sulfate T.S., add 5 ml 1N Hcl, dilute to 500 ml with purified water and mix thoroughly.

4.4.2 Ethoheptazine citrate. The solution shall exhibit maxima absorbances at 263, 257.5 and 251 mu, and minima at 261.5 and 254 mu, when prepared and treated as follows:

Transfer an accurately weighed sample of about 290 mg of the ethoheptazine citrate into a 100 ml glass-stoppered volumetric flask. Add 50 ml N/10 HCl and shake mechanically for 10 minutes. Dilute to volume with N/10 HCl and mix throoughly. Filter through No. 42 Whatman Paper rejecting the first 25 ml of filtrate. Using a suitable recording spectrophotometer, determine the UV maxima and minima, of the solution in a 1 cm quartz cell versus N/10 HCl as the reference solution.

4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 (See 6.3) and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (See 6.1).

5.2.1 Unit of issue. One bottle containing 1000 tablets, as specified, constitutes one unit of issue.

5.2.2 Level A.

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5.2.2.1 Immedaite container. The immediate container shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit packaging. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate packaging. Twelve (12) units of issue shall be packaged as specified in 5.2.9 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. Twenty-four (24) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Twenty-four (24) units of issue shall be packed as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the date of manufacture prefixed by MFD.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

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NOTES: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in paragraph 5.5.3 of PPP-C-186. Date of manufacture shall be shown in lieu of date packed. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE 15°-30°C. (59°-86°F.)"

5.4.3 Exterior container. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186. Date of manufacture shall be shown in lieu of date packed. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE 15°-30°C. (59°-86°F.)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock Number	Item Identification
6505-00-687-7901	ETHOHEPTAZINE CITRATE AND ASPIRIN TABLETS, 100s

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1701

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-E-36700A(DSA-DM) ETHOHEPTAZINE CITRATE AND ASPIRIN TABLETS

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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